Berea Technical College

Department of Engineering Studies
INDUSTRIAL INSTRUMENTS N4

ASSIGNMENT 01

Instructions and Information

Read the questions carefully before answering.

NB! ANSWER ANY TWO QUESTIONS

Drawings to be neat and legible.

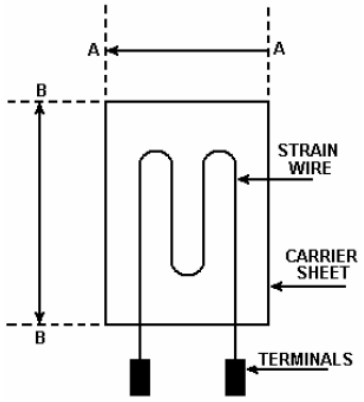
All calculations to be corrected to 3 decimal points

You may use a non – programmable calculator.

All diagrams are to be drawn using lead pencil and labelled using ink (blue or black).

QUESTION 1 **[17]**

1.1 Sturdy the diagram bellow and answer the questions that follow



1.1.1 What is the name of the strain gauge given above? (1)
1.1.2 What type of strain gauge is this ? (1)
1.1.3 Give seven requirements which must be met by resistance of the above mentioned strain gauge. (7)
1.1.4 Show how a strain gauge can be used to measure low pressures. (5)

1.2 Weighing vessel and its contents calls for the installation of load cells.
1.2.1 What is a load cell? (1)
1.2.2 What is a load cell used for? (1)
1.2.3 Is a loadcell a bonded or unbonded type strain gauge? (1)

QUESTION 2 **[27]**

2.1 Make a neat, labelled sketch of an inclined tube manometer. (4)
2.2 What is the manometer in question 1.1 used for? (2)
2.3 The height between the sealed liquid levels in a mercury filled, equal leg manometer is 35 cm, if h = 35 cm, g = 9.8 m/s^2 , p = 13600 kg/m^3 **.** Calculate the appropriate pressure difference indicated by this manometer. (5)
2.4 Explain briefly how the McLeod vacuum pressure gauge operates. (3)
2.5 Make a neat labelled sketch of a Pirani vacuum meter. (4)
2.6 State FOUR factors that will affect the free-end movement of a bourdon tube. (4)
2.7 Determine the angular deflection of a ring balance manometer if the applied pressure difference is 1KPa, the mass of the counter weight is 0.8 Kg, the average radius of the ring is 30 cm and the average radius of the counterweight is 40 cm. the cross sectional area of the ring is 0.00785m^2. (5)

QUESTION 3 **[15]**

3.1 Make a neat labelled sketch of the grid type strain gauge. (4)
3.2 State the differences between bonded type strain gauges and the unbonded type. (2)
3.3 A strain gauge has a nominal resistance of 220 Ω and a gauge factor of 3, it is attached on a beam which is subjected to a stress of 22MPa. If the modulus elasticity is 80, calculate the change is resistance of the gauge of the gauge. (5)
3.4 A dead weight tester contains a piston with the diameter of 45 mm. calculate the mass that must be placed on to the platform to calibrate a pressure gauge at 180 KPa. Ignore the mass of the platform and piston. (4)

QUESTION 4 **[17]**

4.1 What is a weigh bridge used for? (1)

4.2 Make a neat sketch to show how strain gauges are connected to form a weigh bridge. (4)

4.3 Explain how the weight is determined by a weigh bridge (2)

4.4 The piezo-electric effect was first noticed in 1880 by the brothers Pierre and Jacques Curie. By combining their knowledge of pyro-electricity with their understanding of crystal structures and behaviour, the Curie brothers demonstrated the first piezo-electric effect by using crystals of tourmaline, quartz, cane sugar and Rochelle salt.

4.4.1 What is piezo electric effect? (2)
4.4.2 Give TWO applications where piezo electric crystals are use. (2)
4.4.3 Draw a neat sketch of a piezo electric transducer (4)
4.4.4 Give TWO disadvantages of a piezo electric transducer. (2)

TOTAL MARKS = 76